King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam II Term 243

Original exam date: 20 July 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int \frac{e^x}{(e^x-1)(e^x+4)}\,dx =$

(a) $\frac{1}{5}\ln\left|\frac{e^x-1}{e^x+4}\right|+C$

(b) $\ln\left|\frac{e^x-1}{e^x+4}\right|+C$

(c) $\frac{1}{5}\ln\left|\frac{e^x+4}{e^x-1}\right|+C$

(d) $\frac{1}{3}\ln\left|\frac{e^x+4}{e^x-1}\right|+C$

(e) $\frac{1}{3}\ln\left|\frac{e^x-1}{e^x+4}\right|+C$

2. The volume of the solid formed by revolving the region bounded by the graphs of $y=\sqrt{x}$, $y=0$, $x=4$ about the y-axis is

(a) $\frac{64\pi}{3}$

(b) $\frac{128\pi}{3}$

(c) $\frac{64\pi}{5}$

(d) $\frac{128\pi}{5}$

(e) $\frac{128\pi}{7}$

3. The volume of the solid found by revolving the region bounded by the graphs of $y=x^5$, $y=x^2$ about the x-axis is

(a) $\frac{4\pi}{55}$

(b) $\frac{3\pi}{55}$

(c) $\frac{2\pi}{55}$

(d) $\frac{6\pi}{55}$

(e) $\frac{7\pi}{55}$

4. The arc length of the graph of the function $f(x) = \dfrac{x^4}{8} + \dfrac{1}{4x^2}$ over the interval $[1, 2]$ is

(a) $\dfrac{29}{16}$

(b) $\dfrac{31}{16}$

(c) $\dfrac{33}{16}$

(d) $\dfrac{35}{16}$

(e) $\dfrac{27}{16}$

5. If $\dfrac{x^2 - 6x + 2}{x^3 + 2x^2 + x} = \dfrac{A}{x} + \dfrac{B}{x+1} + \dfrac{C}{(x+1)^2}$, then $A + B + C =$

(a) -6

(b) -7

(c) -10

(d) -9

(e) -8

6. $\int \frac{\tan\left(\frac{2}{t}\right)}{t^2}\, dt =$

(a) $\ln\left|\sec\left(\dfrac{2}{t}\right)\right| + C$

(b) $\dfrac{1}{4}\ln\left|\sec\left(\dfrac{2}{t}\right)\right| + C$

(c) $\ln\left|\cos\left(\dfrac{2}{t}\right)\right| + C$

(d) $\dfrac{1}{2}\ln\left|\sec\left(\dfrac{2}{t}\right)\right| + C$

(e) $\dfrac{1}{2}\ln\left|\cos\left(\dfrac{2}{t}\right)\right| + C$

7. $\int \tan^5 x \sec^4 x \, dx =$

(a) $\dfrac{\tan^8 x}{8} + \dfrac{\sec^6 x}{6} + C$

(b) $\dfrac{\sec^8 x}{8} + \dfrac{\sec^6 x}{6} + C$

(c) $\dfrac{\tan^8 x}{8} + \dfrac{\tan^6 x}{6} + C$

(d) $\dfrac{\tan^7 x}{7} + \dfrac{\tan^5 x}{5} + C$

(e) $\dfrac{\sec^7 x}{7} + \dfrac{\sec^5 x}{5} + C$

8. $\int \ln(4 + x^2) \, dx =$

(a) $x \ln(4 + x^2) - 2x + 4 \arctan\left(\dfrac{x}{2}\right) + C$

(b) $x^2 \ln(4 + x^2) - 2x + 4 \arctan\left(\dfrac{x}{2}\right) + C$

(c) $x \ln(4 + x^2) + 2x + 4 \arctan\left(\dfrac{x}{2}\right) + C$

(d) $x \ln(4 + x^2) + 2x + 4 \arctan x + C$

(e) $x \ln(4 + x^2) - 2x + \arctan\left(\dfrac{x}{2}\right) + C$

9. $\int \frac{2x^2}{(4+x^2)^2} \, dx =$

(a) $\dfrac{1}{4} \arctan\left(\dfrac{x}{2}\right) + \dfrac{x}{x^2 + 4} + C$

(b) $\dfrac{1}{2} \arctan\left(\dfrac{x}{2}\right) - \dfrac{x}{x^2 + 4} + C$

(c) $\dfrac{1}{2} \arctan\left(\dfrac{x}{2}\right) + \dfrac{x}{x^2 + 4} + C$

(d) $\dfrac{1}{2} \arctan\left(\dfrac{x}{2}\right) + \dfrac{2x}{x^2 + 4} + C$

(e) $\dfrac{1}{4} \arctan\left(\dfrac{x}{2}\right) - \dfrac{x}{x^2 + 4} + C$

10. $\int \cos^3 x \sin^4 x \, dx =$

(a) $\dfrac{\cos^5 x}{5} - \dfrac{\cos^7 x}{7} + C$

(b) $\dfrac{\cos^4 x}{4} - \dfrac{\cos^6 x}{6} + C$

(c) $\dfrac{\sin^5 x}{5} - \dfrac{\sin^7 x}{7} + C$

(d) $\dfrac{\sin^5 x}{5} - \dfrac{\cos^7 x}{7} + C$

(e) $\dfrac{\sin^4 x}{4} - \dfrac{\sin^6 x}{6} + C$

11. $\int x^5 \ln(3x) \, dx =$

(a) $\dfrac{x^5}{5} \ln(3x) - \dfrac{x^5}{36} + C$

(b) $\dfrac{x^6}{6} \ln x - \dfrac{x^6}{36} + C$

(c) $\dfrac{x^6}{36} \ln(3x) - \dfrac{x^6}{6} + C$

(d) $\dfrac{x^6}{6} \ln(3x) - \dfrac{x^6}{36} + C$

(e) $\dfrac{x^6}{6} \ln(3x) - \dfrac{x^6}{6} + C$

12. The area of the surface generated by revolving the curve $y = \sqrt{4 - x^2}$ on the interval $[-1, 1]$ about the x-axis is

(a) 10π

(b) 8π

(c) 9π

(d) 7π

(e) 11π

13. $\int_0^{\pi/2} \frac{d\theta}{1+\sin\theta+\cos\theta} =$

(a) $\ln 2$

(b) $\ln 3$

(c) $2 \ln 3$

(d) 2

(e) $2 \ln 2$

14. $\int \frac{dx}{(16-x^2)^{3/2}} =$

(a) $\frac{1}{2}\left(\frac{x}{\sqrt{16-x^2}}\right)+C$

(b) $\frac{1}{8}\left(\frac{x}{\sqrt{16-x^2}}\right)+C$

(c) $\frac{x}{\sqrt{16-x^2}}+C$

(d) $\frac{1}{16}\left(\frac{x}{\sqrt{16-x^2}}\right)+C$

(e) $\frac{1}{4}\left(\frac{x}{\sqrt{16-x^2}}\right)+C$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$u = e^x$, $\frac{1}{(u-1)(u+4)} = \frac{1}{5}(\frac{1}{u-1} - \frac{1}{u+4})$, giving $\frac{1}{5}\ln\lvert\frac{e^x-1}{e^x+4}\rvert$.
2	**(d)**	Shells: $2\pi\int_0^4 x^{3/2}dx = 2\pi\cdot\frac{2}{5}\cdot 32 = \frac{128\pi}{5}$.
3	**(d)**	Washers: $\pi\int_0^1(x^4 - x^{10})dx = \pi(\frac{1}{5} - \frac{1}{11}) = \frac{6\pi}{55}$.
4	**(c)**	$\sqrt{1+f'^2} = \frac{x^3}{2} + \frac{1}{2x^3}$; $L = [\frac{x^4}{8} - \frac{1}{4x^2}]_1^2 = \frac{33}{16}$.
5	**(e)**	$A = 2$, $C = -9$ (plug $x = 0, -1$), $B = 1 - A = -1$; sum $= -8$.
6	**(e)**	$u = \frac{2}{t}$, $\frac{dt}{t^2} = -\frac{1}{2}du$: $\frac{1}{2}\ln\lvert\cos\frac{2}{t}\rvert + C$.
7	**(c)**	$u = \tan x$: $\int(u^5 + u^7)du = \frac{\tan^8 x}{8} + \frac{\tan^6 x}{6} + C$.
8	**(a)**	Parts, then $\frac{2x^2}{4+x^2} = 2 - \frac{8}{4+x^2}$: $x\ln(4+x^2) - 2x + 4\arctan\frac{x}{2} + C$.
9	**(b)**	$x = 2\tan\theta$: $\int\sin^2\theta\,d\theta = \frac{\theta}{2} - \frac{\sin 2\theta}{4}$, giving $\frac{1}{2}\arctan\frac{x}{2} - \frac{x}{x^2+4} + C$.
10	**(c)**	$u = \sin x$: $\int(u^4 - u^6)du = \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + C$.
11	**(d)**	Parts with $u = \ln 3x$, $dv = x^5dx$: $\frac{x^6}{6}\ln(3x) - \frac{x^6}{36} + C$.
12	**(b)**	$y\sqrt{1+y'^2} = 2$, so $S = 2\pi\int_{-1}^1 2\,dx = 8\pi$.
13	**(a)**	$u = \tan\frac{\theta}{2}$: integrand becomes $\frac{du}{1+u}$, so $\int_0^1\frac{du}{1+u} = \ln 2$.
14	**(d)**	$x = 4\sin\theta$: $\frac{1}{16}\int\sec^2\theta\,d\theta = \frac{1}{16}\frac{x}{\sqrt{16-x^2}} + C$.